HCR 3.0 Probe Maker v1.0

Create probes for multiplexed, fluorescent *in situ* hybridization

IDT oPool submission format

Copy and Paste the lines below into an XLS or XLSX file

Pool name	Sequence
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAAAACTTTGATAACGCAGCCTTCGCTG
B1_mylt_23_Delay0	TTGATGGCAGATTTTACGATGCCACTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAATATCTATTCATCTATCTATCTGTCC
B1_mylt_23_Delay0	CATTTGACCTCAACTGTACACAATATAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAAACAAGACTCATGCACTCGGCAAATG
B1_mylt_23_Delay0	TTTCTATCACATTCGTCACAGTTGTTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAACTATAACTGCCATTCTCAATGCATG
B1_mylt_23_Delay0	TTTCACAATAATGACTCTTTGAAAGTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAACTTCGACACCTCTGCTTCAGGTTTA
B1_mylt_23_Delay0	ATTATTCATCAAGATTTCAAGAGCCTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAAATATCAGCAGTTGAATTAGGTAATC
B1_mylt_23_Delay0	TATAGCTCTTACAGCTACGCTTCATTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAAACGAGACCTTAGCACTATTCCGAAG
B1_mylt_23_Delay0	TGAAAACATTGATATTTCTTCTTAGTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAAGCCTAATTTGTGCATAATGTCAATC
B1_mylt_23_Delay0	GACAGATGTAACTGGATCATGCTTGTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAATCGTGGCCTACTTGACCAGGGACAG
B1_mylt_23_Delay0	TAGGTGAGAGGTCTCAAATTAACATTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAACGCACAATGTTGCTCAAGTAAATGC
B1_mylt_23_Delay0	CTTAAAACTATACCATGTTTAGCATTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAACTTCAAGGGACATGACCCAGTCGTA
B1_mylt_23_Delay0	CGAATTCTCTCGGGATGTGTTCCGCTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAAGTCTAGGTACTTCACTCGGCCGTTG
B1_mylt_23_Delay0	TGGTAACCCATATTGAGGTAGAGACTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAATAGGCATCATAGCGATACAAAAGCT
B1_mylt_23_Delay0	ACCTTGTCATAGTATGGGCCACCATTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAATCTCCAAAGCTTCCTACTACCTTTC
B1_mylt_23_Delay0	AAAAACTTTTTGTAATGGCCACGTCTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAAGTTTATCAGTGTAGTGATTATACTC
B1_mylt_23_Delay0	AAAAATTAGCAAGACTGGTGGCCAATAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAAACCAGATTCAGAAACTGCCTTGTTT
B1_mylt_23_Delay0	TCCTTTGGCTGTTTCCTCACACCATTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAATTTTGTGAAACAGAATTAAGAAGTC
B1_mylt_23_Delay0	TGTAACTGGCAGAAGAGAATAGCCTTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAATCAACATGTTAGATGTTGGCACAGC
B1_mylt_23_Delay0	AAAGTTGATCCCTCTGCTGTTTGAATAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAAATAATAATGGAATAGGGCAAACAAC
B1_mylt_23_Delay0	GGGTATTACTGCAGATATTCGTCTCTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAACATGGCAAGCATCCTCTCAGCTTGA
B1_mylt_23_Delay0	AAAGCAAATAACAGTATTTTAGGCATAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAACACACGATAATGCCCTCTTTGATAA
B1_mylt_23_Delay0	GTCGAGACTCCTGATCAAAAACCATTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAATATAAGTGTCCAAATCACAAGTACA
B1_mylt_23_Delay0	AACATCAACCTCACACTGAGATTAGTAGAAGAGTCTTCCTTTACG
B1_mylt_23_Delay0	GAGGAGGGCAGCAAACGGAATGCCAAACAACTTTCAATTTCCAAG
B1_mylt_23_Delay0	CTTCTTATTGCTTAGTGCACCCAGGTAGAAGAGTCTTCCTTTACG

Below are the hybridizing sequences and where they align to the cDNA

Pair	Coordinate	Hybridization	Coordinate	Coordinate	Hybridization	Coordinate
23	86	CTTCTTATTGCTTAGTGCACCCAGG	61	59	TGCCAAACAACTTTCAATTTCCAAG	34
22	141	AACATCAACCTCACACTGAGATTAG	116	114	TATAAGTGTCCAAATCACAAGTACA	89
21	196	GTCGAGACTCCTGATCAAAAACCAT	171	169	CACACGATAATGCCCTCTTTGATAA	144
20	251	AAAGCAAATAACAGTATTTTAGGCA	226	224	CATGGCAAGCATCCTCTCAGCTTGA	199
19	306	GGGTATTACTGCAGATATTCGTCTC	281	279	ATAATAATGGAATAGGGCAAACAAC	254
18	361	AAAGTTGATCCCTCTGCTGTTTGAA	336	334	TCAACATGTTAGATGTTGGCACAGC	309
17	416	TGTAACTGGCAGAAGAGAATAGCCT	391	389	TTTTGTGAAACAGAATTAAGAAGTC	364
16	483	TCCTTTGGCTGTTTCCTCACACCAT	458	456	ACCAGATTCAGAAACTGCCTTGTTT	431
15	538	AAAAATTAGCAAGACTGGTGGCCAA	513	511	GTTTATCAGTGTAGTGATTATACTC	486
14	596	AAAAACTTTTTGTAATGGCCACGTC	571	569	TCTCCAAAGCTTCCTACTACCTTTC	544
13	659	ACCTTGTCATAGTATGGGCCACCAT	634	632	TAGGCATCATAGCGATACAAAAGCT	607
12	714	TGGTAACCCATATTGAGGTAGAGAC	689	687	GTCTAGGTACTTCACTCGGCCGTTG	662
11	769	CGAATTCTCTCGGGATGTGTTCCGC	744	742	CTTCAAGGGACATGACCCAGTCGTA	717
10	824	CTTAAAACTATACCATGTTTAGCAT	799	797	CGCACAATGTTGCTCAAGTAAATGC	772
9	881	TAGGTGAGAGGTCTCAAATTAACAT	856	854	TCGTGGCCTACTTGACCAGGGACAG	829
8	936	GACAGATGTAACTGGATCATGCTTG	911	909	GCCTAATTTGTGCATAATGTCAATC	884
7	991	TGAAAACATTGATATTTCTTCTTAG	966	964	ACGAGACCTTAGCACTATTCCGAAG	939
6	1046	TATAGCTCTTACAGCTACGCTTCAT	1021	1019	ATATCAGCAGTTGAATTAGGTAATC	994
5	1198	ATTATTCATCAAGATTTCAAGAGCC	1173	1171	CTTCGACACCTCTGCTTCAGGTTTA	1146
4	1253	TTTCACAATAATGACTCTTTGAAAG	1228	1226	CTATAACTGCCATTCTCAATGCATG	1201
3	1395	TTTCTATCACATTCGTCACAGTTGT	1370	1368	ACAAGACTCATGCACTCGGCAAATG	1343
2	1450	CATTTGACCTCAACTGTACACAATA	1425	1423	TATCTATTCATCTATCTATCTGTCC	1398
1	1510	TTGATGGCAGATTTTACGATGCCAC	1485	1483	AACTTTGATAACGCAGCCTTCGCTG	1458

Here are the probes in figure format

Pair	Initiator	Spacer	Hybridzation	Hybridzation	Spacer	Initiator
1	GAGGAGGGCAGCAAACGG	AA	AACTTTGATAACGCAGCCTTCGCTG	TTGATGGCAGATTTTACGATGCCAC	TA	GAAGAGTCTTCCTTTACG
2	GAGGAGGGCAGCAAACGG	AA	TATCTATTCATCTATCTATCTGTCC	CATTTGACCTCAACTGTACACAATA	TA	GAAGAGTCTTCCTTTACG
3	GAGGAGGGCAGCAAACGG	AA	ACAAGACTCATGCACTCGGCAAATG	TTTCTATCACATTCGTCACAGTTGT	TA	GAAGAGTCTTCCTTTACG
4	GAGGAGGGCAGCAAACGG	AA	CTATAACTGCCATTCTCAATGCATG	TTTCACAATAATGACTCTTTGAAAG	TA	GAAGAGTCTTCCTTTACG
5	GAGGAGGGCAGCAAACGG	AA	CTTCGACACCTCTGCTTCAGGTTTA	ATTATTCATCAAGATTTCAAGAGCC	TA	GAAGAGTCTTCCTTTACG
6	GAGGAGGGCAGCAAACGG	AA	ATATCAGCAGTTGAATTAGGTAATC	TATAGCTCTTACAGCTACGCTTCAT	TA	GAAGAGTCTTCCTTTACG
7	GAGGAGGGCAGCAAACGG	AA	ACGAGACCTTAGCACTATTCCGAAG	TGAAAACATTGATATTTCTTCTTAG	TA	GAAGAGTCTTCCTTTACG
8	GAGGAGGGCAGCAAACGG	AA	GCCTAATTTGTGCATAATGTCAATC	GACAGATGTAACTGGATCATGCTTG	TA	GAAGAGTCTTCCTTTACG
9	GAGGAGGGCAGCAAACGG	AA	TCGTGGCCTACTTGACCAGGGACAG	TAGGTGAGAGGTCTCAAATTAACAT	TA	GAAGAGTCTTCCTTTACG
10	GAGGAGGGCAGCAAACGG	AA	CGCACAATGTTGCTCAAGTAAATGC	CTTAAAACTATACCATGTTTAGCAT	TA	GAAGAGTCTTCCTTTACG
11	GAGGAGGGCAGCAAACGG	AA	CTTCAAGGGACATGACCCAGTCGTA	CGAATTCTCTCGGGATGTGTTCCGC	TA	GAAGAGTCTTCCTTTACG
12	GAGGAGGGCAGCAAACGG	AA	GTCTAGGTACTTCACTCGGCCGTTG	TGGTAACCCATATTGAGGTAGAGAC	TA	GAAGAGTCTTCCTTTACG
13	GAGGAGGGCAGCAAACGG	AA	TAGGCATCATAGCGATACAAAAGCT	ACCTTGTCATAGTATGGGCCACCAT	TA	GAAGAGTCTTCCTTTACG
14	GAGGAGGGCAGCAAACGG	AA	TCTCCAAAGCTTCCTACTACCTTTC	AAAAACTTTTTGTAATGGCCACGTC	TA	GAAGAGTCTTCCTTTACG
15	GAGGAGGGCAGCAAACGG	AA	GTTTATCAGTGTAGTGATTATACTC	AAAAATTAGCAAGACTGGTGGCCAA	TA	GAAGAGTCTTCCTTTACG
16	GAGGAGGGCAGCAAACGG	AA	ACCAGATTCAGAAACTGCCTTGTTT	TCCTTTGGCTGTTTCCTCACACCAT	TA	GAAGAGTCTTCCTTTACG
17	GAGGAGGGCAGCAAACGG	AA	TTTTGTGAAACAGAATTAAGAAGTC	TGTAACTGGCAGAAGAGAATAGCCT	TA	GAAGAGTCTTCCTTTACG
18	GAGGAGGGCAGCAAACGG	AA	TCAACATGTTAGATGTTGGCACAGC	AAAGTTGATCCCTCTGCTGTTTGAA	TA	GAAGAGTCTTCCTTTACG
19	GAGGAGGGCAGCAAACGG	AA	ATAATAATGGAATAGGGCAAACAAC	GGGTATTACTGCAGATATTCGTCTC	TA	GAAGAGTCTTCCTTTACG
20	GAGGAGGGCAGCAAACGG	AA	CATGGCAAGCATCCTCTCAGCTTGA	AAAGCAAATAACAGTATTTTAGGCA	TA	GAAGAGTCTTCCTTTACG
21	GAGGAGGGCAGCAAACGG	AA	CACACGATAATGCCCTCTTTGATAA	GTCGAGACTCCTGATCAAAAACCAT	TA	GAAGAGTCTTCCTTTACG
22	GAGGAGGGCAGCAAACGG	AA	TATAAGTGTCCAAATCACAAGTACA	AACATCAACCTCACACTGAGATTAG	TA	GAAGAGTCTTCCTTTACG

23 GAGGAGGGCAGCAAACGG AA TGCCAAACAACTTTCAATTTCCAAG CTTCTTATTGCTTAGTGCACCCAGG TA GAAGAGTCTTCCTTTACG

Below is the in-place location of the probe pairs in the cDNA sense strand

```
>mylt Sense Strand
nnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnCTTGGAAATTGAAAGTTGTTTGGCAn
nCCTGGGTGCACTAAGCAATAAGAAGnnnTGTACTTGTGATTTGGACACTTATAnnCTAA
TCTCAGTGTGAGGTTGATGTTnnnTTATCAAAGAGGGCATTATCGTGTGnnATGGTTTTT
GATCAGGAGTCTCGACnnnTCAAGCTGAGAGGATGCTTGCCATGnnTGCCTAAAATACTG
TTATTTGCTTTnnnGTTGTTTGCCCTATTCCATTATTATnnGAGACGAATATCTGCAGTA
ATACCCnnnGCTGTGCCAACATCTAACATGTTGAnnTTCAAACAGCAGAGGGATCAACTT
TnnnGACTTCTTAATTCTGTTTCACAAAAnnAGGCTATTCTCTTCTGCCAGTTACAnnnn
nnnnnnnnnnnAAACAAGGCAGTTTCTGAATCTGGTnnATGGTGTGAGGAAACAGCCAAA
GGAnnnGAGTATAATCACTACACTGATAAACnnTTGGCCACCAGTCTTGCTAATTTTTnn
nnnnGAAAGGTAGTAGGAAGCTTTGGAGAnnGACGTGGCCATTACAAAAAGTTTTTnnnn
nnnnnnnAGCTTTTGTATCGCTATGATGCCTAnnATGGTGGCCCATACTATGACAAGGTn
nnCAACGGCCGAGTGAAGTACCTAGACnnGTCTCTACCTCAATATGGGTTACCAnnnTAC
GACTGGGTCATGTCCCTTGAAGnnGCGGAACACATCCCGAGAGAATTCGnnnGCATTTAC
TTGAGCAACATTGTGCGnnATGCTAAACATGGTATAGTTTTAAGnnnnnCTGTCCCTGGT
CAAGTAGGCCACGAnnATGTTAATTTGAGACCTCTCACCTAnnnGATTGACATTATGCAC
AAATTAGGCnnCAAGCATGATCCAGTTACATCTGTCnnnCTTCGGAATAGTGCTAAGGTC
TCGTnnCTAAGAAGAAATATCAATGTTTTCAnnnGATTACCTAATTCAACTGCTGATATn
nATGAAGCGTAGCTGTAAGAGCTATAnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnn
nnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnn
nnnnnnTAAACCTGAAGCAGAGGTGTCGAAGnnGGCTCTTGAAATCTTGATGAATAATnn
nCATGCATTGAGAATGGCAGTTATAGnnCTTTCAAAGAGTCATTATTGTGAAAnnnnnnn
nnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnnn
nnnnnnnnnnnnnnnnnnnnnnnCATTTGCCGAGTGCATGAGTCTTGTnnACAACTGTGA
CGAATGTGATAGAAAnnnGGACAGATAGATAGATGAATAGATAnnTATTGTGTACAGTTG
AGGTCAAATGnnnnnnnnCAGCGAAGGCTGCGTTATCAAAGTTnnGTGGCATCGTAAAAT
CTGCCATCAA
```

Below is the anti-sense sequence used to create your probes

```
>mylt Anti-Sense Strand
TTGATGGCAGATTTTACGATGCCACATAACTTTGATAACGCAGCCTTCGCTGTATGAATA
CATTTGACCTCAACTGTACACAATACCTATCTATTCATCTATCTATCTGTCCATCTTTCT
ATCACATTCGTCACAGTTGTTCACAAGACTCATGCACTCGGCAAATGATTGAAACTCCTG
TGCTATGTAATTATCTGATGTTCTATCCAACATCATAATAAAACACTTGATTCTGCGGTG
ACATATCACAAAAAAATTTTCACAATAATGACTCTTTGAAAGGCCTATAACTGCCATTCT
CAATGCATGGACATTATTCATCAAGATTTCAAGAGCCAACTTCGACACCTCTGCTTCAGG
TTTATATCATCAGTCAAAATAACTTTCATCATTAAAACGGGACAAATCTTTAGTTTTTGG
GTAAAATTTATTGACAACAAAATACAAAATCAAACATGTATATATATAGCTCTTACAGCT
ACGCTTCATTGATATCAGCAGTTGAATTAGGTAATCTCCTGAAAACATTGATATTTCTTC
TTAGCCACGAGACCTTAGCACTATTCCGAAGGTTGACAGATGTAACTGGATCATGCTTGA
AGCCTAATTTGTGCATAATGTCAATCACATAGGTGAGAGGTCTCAAATTAACATGCTCGT
GGCCTACTTGACCAGGGACAGCCCAACTTAAAACTATACCATGTTTAGCATGTCGCACAA
TGTTGCTCAAGTAAATGCTTTCGAATTCTCTCGGGATGTGTTCCGCCACTTCAAGGGACA
TGACCCAGTCGTAAAGTGGTAACCCATATTGAGGTAGAGACAAGTCTAGGTACTTCACTC
GGCCGTTGCTAACCTTGTCATAGTATGGGCCACCATCATAGGCATCATAGCGATACAAAA
GCTTCGTCTTGTCAAAAAACTTTTTGTAATGGCCACGTCCATCTCCAAAGCTTCCTACTA
CCTTTCCAGCAAAAAAATTAGCAAGACTGGTGGCCAAACGTTTATCAGTGTAGTGATTAT
ACTCATTTCCTTTGGCTGTTTCCTCACACCATCCACCAGATTCAGAAACTGCCTTGTTTT
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CTTTTTGATGTTTTTGTAACTGGCAGAAGAGAATAGCCTTCTTTTGTGAAACAGAATTAA
GAAGTCTGGAAAGTTGATCCCTCTGCTGTTTGAACCTCAACATGTTAGATGTTGGCACAG
CCCTGGGTATTACTGCAGATATTCGTCTCGGATAATAATGGAATAGGGCAAACAACAGGA
AAGCAAATAACAGTATTTTAGGCAACCATGGCAAGCATCCTCTCAGCTTGACAAGTCGAG
ACTCCTGATCAAAAACCATCTCACACGATAATGCCCTCTTTGATAATCAAACATCAACCT
CACACTGAGATTAGGCTATAAGTGTCCAAATCACAAGTACACAGCTTCTTATTGCTTAGT
GCACCCAGGCATGCCAAACAACTTTCAATTTCCAAGTCCCCATGTACTCTGCGTTGATAC
CACTGCTTAG
```

These are the settings used to make your probes

Parameter	Setting
Date Run	2023-08-03
5' Bases Skipped	0
Probe Set Chosen	Default
Max A/T homopolymer length	5
Max C/G homopolymer length	4
Upper Probe CG% limit	55%
Lower Probe CG% limit	35%
Run with BLAST?	n

Software written by:

Ryan W. Null with M. Desmond Ramirez, Dennis Sun, and B. Duygu Özpolat.

Cite Us!

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For more information visit GitHub

References:

For HCR3.0 methodology details see: Choi et al. 2018 Development and Molecular Instruments

For expanded amplifier/initiator details see:
Wang et al. 2020 BioRxiv and Tsuneoka et al. 2020 Front. Mol. Neurosci.

Washington University in St. Louis

Send us an email if you have found a bug, or if you have suggestions to share.